Final Lab Write Up
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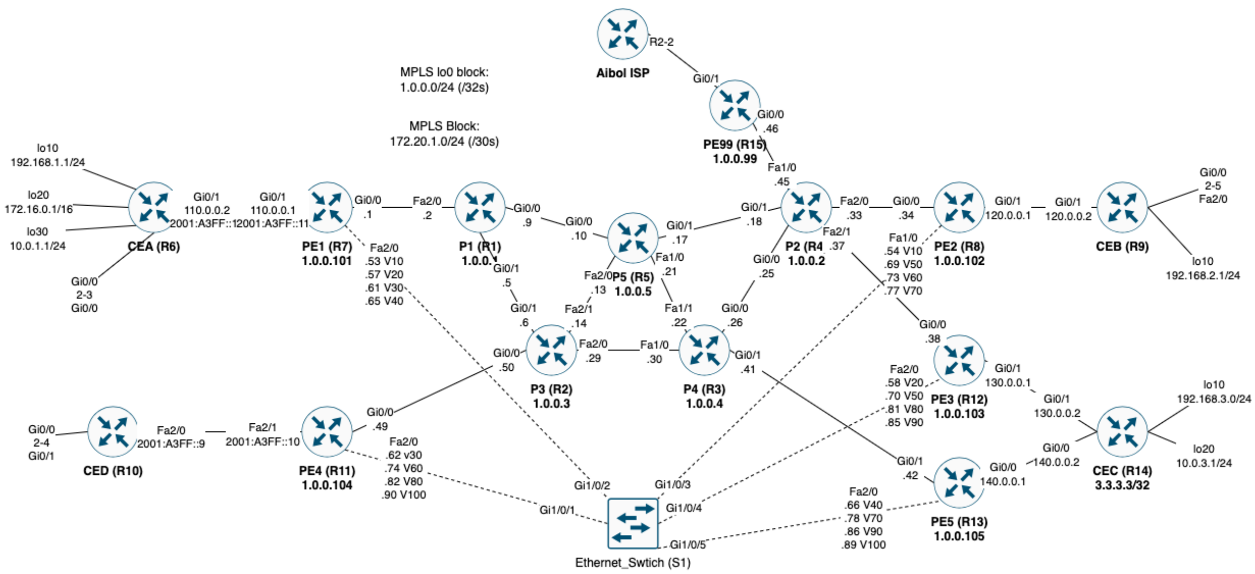
Enterprise Networks
5/6/25

Network Diagram:



IP Addressing:

For MPLS backbone, I used the 172.20.1.0/24 network for P2P links sectioned into /30s

MPLS & Underlay IGP:

Initially I configured OSPF for all PE facing P interfaces and all interfaces on P devices. Each device has a loopback for easier view.

OSPF:

OSPF configuration (Example of PE1):

```
router ospf 1
mpls ldp autoconfig
router-id 1.0.0.101
```

```
network 1.0.0.101 0.0.0.0 area 0
network 172.20.1.0 0.0.0.255 area 0
!
```

LDP:

As you can see, to enable LDP OSPF has an autoconfig command that enables LDP without having to enter “mpls ip” on every interface.

Proof of LDP (Example of P5):

```
P5#show mpls interfaces
Interface              IP            Tunnel   BGP Static Operational
GigabitEthernet0/0     Yes (ldp)     No       No  No     Yes
GigabitEthernet0/1     Yes (ldp)     No       No  No     Yes
FastEthernet1/0        Yes (ldp)     No       No  No     Yes
FastEthernet2/0        Yes (ldp)     No       No  No     Yes
P5#
```

MPLS TE:

All P routers config:

```
P Config:
mpls traffic-eng tunnels
router ospf 1
mpls traffic-eng router-id loopback 0
mpls traffic-eng area 0
*** FOR EVERY INTERFACE WITH MPLS: ***
interface gi0/1
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
```

All PE routers config:

```
mpls traffic-eng tunnels
router ospf 1
mpls traffic-eng router-id loopback 0
mpls traffic-eng area 0
interface gi0/1
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
```

PE1 -> PE2 (50MBs):

```
interface Tunnel1
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
```

```
tunnel destination 1.0.0.102
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 5000
tunnel mpls traffic-eng path-option 1 explicit name PE1_PE2 lockdown
tunnel mpls traffic-eng path-option 2 explicit name PE1_PE2_backup
no routing dynamic
no shut
!
ip explicit-path name PE1_PE2 enable
next-address 1.0.0.1
next-address 1.0.0.3
next-address 1.0.0.5
next-address 1.0.0.2
next-address 1.0.0.102
!
ip explicit-path name PE1_PE2_backup enable
next-address 1.0.0.1
next-address 1.0.0.5
next-address 1.0.0.4
next-address 1.0.0.2
next-address 1.0.0.102
!
```

PE1 -> PE3 (50MBs):

```
interface Tunnel2
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
tunnel destination 1.0.0.103
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 5000
tunnel mpls traffic-eng path-option 1 explicit name PE1_PE3 lockdown
tunnel mpls traffic-eng path-option 2 dynamic
no routing dynamic
no shut
!
ip explicit-path name PE1_PE3 enable
next-address 1.0.0.1
next-address 1.0.0.3
next-address 1.0.0.4
next-address 1.0.0.2
next-address 1.0.0.103
```

PE1 -> PE5 (50MBs):

```
interface Tunnel3
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
tunnel destination 1.0.0.105
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 5000
tunnel mpls traffic-eng path-option 1 explicit name PE1_PE5 lockdown
tunnel mpls traffic-eng path-option 2 dynamic
```

```
no routing dynamic
no shut
!
ip explicit-path name PE1_PE5 enable
next-address 1.0.0.1
next-address 1.0.0.5
next-address 1.0.0.4
next-address 1.0.0.105
```

PE2 -> PE1 (30MBs):

```
interface Tunnel1
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
tunnel destination 1.0.0.101
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 3000
tunnel mpls traffic-eng path-option 1 explicit name PE2_PE1 lockdown
tunnel mpls traffic-eng path-option 2 explicit name PE2_PE1_backup
no routing dynamic
no shut
!
ip explicit-path name PE2_PE1 enable
next-address 1.0.0.2
next-address 1.0.0.5
next-address 1.0.0.3
next-address 1.0.0.1
next-address 1.0.0.101
!
ip explicit-path name PE2_PE1_backup enable
next-address 1.0.0.2
next-address 1.0.0.4
next-address 1.0.0.5
next-address 1.0.0.1
next-address 1.0.0.101
!
```

PE2 -> PE4 (30MBs):

```
interface Tunnel2
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
tunnel destination 1.0.0.104
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 3000
tunnel mpls traffic-eng path-option 1 explicit name PE2_PE4 lockdown
tunnel mpls traffic-eng path-option 2 dynamic
no routing dynamic
no shut
!
ip explicit-path name PE2_PE4 enable
next-address 1.0.0.2
next-address 1.0.0.5
next-address 1.0.0.1
```

```
next-address 1.0.0.3
next-address 1.0.0.104
!
```

PE2 -> PE5 (30MBs):

```
interface Tunnel3
ip unnumbered Loopback0
tunnel mode mpls traffic-eng
tunnel destination 1.0.0.105
tunnel mpls traffic-eng autoroute announce
tunnel mpls traffic-eng priority 7 7
tunnel mpls traffic-eng bandwidth 3000
tunnel mpls traffic-eng path-option 1 explicit name PE2_PE5 lockdown
tunnel mpls traffic-eng path-option 2 dynamic
no routing dynamic
no shut
!
ip explicit-path name PE2_PE5 enable
next-address 1.0.0.2
next-address 1.0.0.5
next-address 1.0.0.4
next-address 1.0.0.105
!
```

Ethernet Switch Connectivity

To connect all of the PEs together as a last resort, connections were made in the ethernet switch. To connect them together, I made all switchports trunks and created sub interfaces on all devices that connect to every other respective PE.

Here is an example of PE1, as the other PE devices had the same exact configuration except for the actual IPs and interfaces:

```
interface FastEthernet2/0.10
encapsulation dot1Q 10
ip address 172.20.1.53 255.255.255.252
ip ospf cost 10000
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
!
interface FastEthernet2/0.20
encapsulation dot1Q 20
ip address 172.20.1.57 255.255.255.252
ip ospf cost 10000
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
!
interface FastEthernet2/0.30
encapsulation dot1Q 30
ip address 172.20.1.61 255.255.255.252
ip ospf cost 10000
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
!
```

```
interface FastEthernet2/0.40
encapsulation dot1Q 40
ip address 172.20.1.65 255.255.255.252
ip ospf cost 10000
mpls traffic-eng tunnels
ip rsvp bandwidth 100000 50000
!
```

IPv4 Tunnels + Routing:

IPv4 IPSec GRE:

Tunnels were created between each CE router with v4 customers. This was done with IPSec for encryption/authentication and used a GRE to pass actual traffic. After that was completed OSPF was ran between different CEs, following specifications for area 0 and area 100:

CEA-CEB:

```
CEA-CEB:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEACEB address 120.0.0.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel1
ip address 10.10.10.1 255.255.255.0
tunnel source 110.0.0.2
tunnel destination 120.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
!
router ospf 10
router-id 1.1.1.1
network 10.0.1.0 0.0.0.255 area 0
network 10.10.10.0 0.0.0.255 area 0
network 172.16.0.0 0.0.255.255 area 0
!
```

CEA-CEC:

```
CEA-CEC:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
```

```
crypto isakmp key CEACEC address 3.3.3.3
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel2
ip address 20.20.20.1 255.255.255.0
tunnel source 110.0.0.2
tunnel destination 3.3.3.3
tunnel protection ipsec profile GREOVERIPSEC
!
```

CEB-CEA:

```
CEB-CEA:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEACEB address 110.0.0.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel1
ip address 10.10.10.2 255.255.255.0
tunnel source 120.0.0.2
tunnel destination 110.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
!
router ospf 10
router-id 2.2.2.2
network 10.10.10.0 0.0.0.255 area 0
network 192.168.2.0 0.0.0.255 area 0
!
```

CEB-CEC:

```
CEB-CEC:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 3.3.3.3
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel2
ip address 30.30.30.1 255.255.255.0
```

```
tunnel source 120.0.0.2
tunnel destination 3.3.3.3
tunnel protection ipsec profile GREOVERIPSEC
!
router ospf 10
network 30.30.30.0 0.0.0.255 area 100
!
```

CEC-CEA:

```
CEC-CEA:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEACEC address 110.0.0.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel1
ip address 20.20.20.2 255.255.255.0
tunnel source 3.3.3.3
tunnel destination 110.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
!
router ospf 10
router-id 3.3.3.3
network 20.20.20.0 0.0.0.255 area 0
network 192.168.3.0 0.0.0.255 area 0
network 10.0.3.1 0.0.0.255 area 0
!
```

CEC-CEB:

```
CEC-CEB:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 120.0.0.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel2
ip address 30.30.30.2 255.255.255.0
tunnel source 3.3.3.3
tunnel destination 120.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
!
router ospf 10
```

```
network 30.30.30.0 0.0.0.255 area 100
!
```

Global Connectivity and NAT:

We now have connectivity between the customer, but what about the internet? To achieve this, there was PAT configured on all CE routers towards the PE, where the PE has a default route to PE99. From there connectivity is handed off via one more PAT to the internet switch.

NAT config:

```
ip nat inside source list NAT interface GigabitEthernet0/1 overload
ip route 0.0.0.0 0.0.0.0 110.0.0.1
!
ip access-list extended NAT
permit ip 192.168.1.0 0.0.0.255 any
permit ip 172.16.0.0 0.0.255.255 any
permit ip 10.0.1.0 0.0.0.255 any
!
interface GigabitEthernet0/1
ip nat outside
!
interface LoopbackX
ip nat inside
!
```

For global connectivity, I made sure to set up RSVP tunnels between PE99 and the respective PEs to get connectivity via tunneling and set static routes to point PE99 to my sites and the PEs having that default route out could get connection back to PE99. CEs also had a default route out as we would see as regular behavior from a customer router.

IPv6 PD, DHCPv6 Stateful/Stateless and SLAAC:

For prefix delegation, I made a CUSTOMER pool that pd clients would grab from and i could assign prefixes on customer router interfaces to get IPv6 IPs whether it was DHCPv6 Stateless/Stateful or SLAAC

Prefix Delegation:

```
int Fa2/1
*** INTERFACES TO CE ROUTERS***
ipv6 dhcp server CUSTOMERS
!
ipv6 dhcp pool CUSTOMERS
```

```
prefix-delegation pool GLOBAL_POOL
ipv6 local pool GLOBAL_POOL 2001:A3::/38 48
```

DHCP Stateful:

```
ipv6 dhcp pool STATEFUL
address prefix 2001:A3:1:2::2/64
dns-server 2001:4860:4860::8888
domain-name NETWORKLESSONS.LOCAL
!
interface GigabitEthernet0/0
ipv6 address ISP_PREFIX ::2:0:0:0:2/64
ipv6 nd managed-config-flag
ipv6 dhcp server STATEFUL
!
interface Tunnel3
ipv6 dhcp client pd ISP_PREFIX
!
```

DHCP Stateless:

```
ipv6 dhcp pool STATELESS
dns-server 2001:4860:4860::8888
domain-name NETWORKLESSONS.LOCAL
!
interface GigabitEthernet0/0
ipv6 address ISP_PREFIX ::1/64
ipv6 nd other-config-flag
ipv6 dhcp server STATELESS
!
interface Tunnel4
ipv6 dhcp client pd ISP_PREFIX
!
```

IPv6 IPSec GRE Tunnels + Routing:

To achieve full IPv6 connectivity and Routing, I first created IPSec GRE tunnels between PE1, PE4 and CEB. This config was similar to the IPSec GRE v4 tunnels, but the actual tunnel IP was IPv6. OSPFv3 was very simple to set up, exposing networks to it based on interface/tunnel to achieve site to site connectivity.

IPSec GRE Tunnels:

PE1-CEB & CEB-PE1

```
PE1-CEB:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 120.0.0.2
```

```
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel5
ipv6 enable
ipv6 address 2001:a3ff::0/127
tunnel source 1.0.0.101
tunnel destination 120.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
!

CEB-PE1:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 1.0.0.101
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel3
ipv6 enable
ipv6 address 2001:a3ff::1/127
tunnel source 120.0.0.2
tunnel destination 1.0.0.101
tunnel protection ipsec profile GREOVERIPSEC
!
```

PE4-CEB & CEB-PE4:

```
PE4-CEB:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 120.0.0.2
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel3
ipv6 enable
ipv6 address 2001:a3ff::2/127
tunnel source 1.0.0.104
tunnel destination 120.0.0.2
tunnel protection ipsec profile GREOVERIPSEC
```

```
!


CEB-PE4:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 1.0.0.104
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel4
ipv6 enable
ipv6 address 2001:a3ff::3/127
tunnel source 120.0.0.2
tunnel destination 1.0.0.104
tunnel protection ipsec profile GREOVERIPSEC
!
```

PE1-PE4 & PE4-PE1:

```
PE1-PE4:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 1.0.0.104
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
set transform-set MYTRANSFORMSET
!
interface Tunnel6
ipv6 enable
ipv6 address 2001:a3ff::4/127
tunnel source 1.0.0.101
tunnel destination 1.0.0.104
tunnel protection ipsec profile GREOVERIPSEC
!


PE4-PE1:
crypto isakmp policy 1
encr aes
authentication pre-share
group 2
crypto isakmp key CEBCEC address 1.0.0.101
!
crypto ipsec transform-set MYTRANSFORMSET esp-aes esp-sha-hmac
!
crypto ipsec profile GREOVERIPSEC
```

```
set transform-set MYTRANSFORMSET
!
interface Tunnel4
ipv6 enable
ipv6 address 2001:a3ff::5/127
tunnel source 1.0.0.104
tunnel destination 1.0.0.101
tunnel protection ipsec profile GREOVERIPSEC
!
```

OSPFv3:

```
ipv6 router ospf 1
*** FOR RESPECTIVE TUNNELS ***
int Tunnel 3
ipv6 ospf 1 area 0
```